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SOFTWARE ENGINEERING

ARCHITECTURE

Architecture Selection: Layered Architecture for the Municipal Infrastructure Damage Reporting App System.

Reasons

The system consists of three horizontal layers: Presentation Layer (Citizen Reporting component), Business Layer (Damage Processing, GPS/GIS, and Ticket Dispatch components), and Data Layer (Municipal Database component). Each of these layers communicates only with the adjacent layer (below or above).

1. Well-defined separation of concerns for a moderate-size civic application: citizen damage reports follow an obvious linear pipeline, from submission to validation/processing, to GPS extraction, to ticket dispatch, to data storage. The pipeline can be mapped on a layered architecture easily. Each concern of capturing the input, processing the business rules, dealing with locations and storing data is kept separate and independently modifiable (by changing the respective layer) without making the city adopt the microservices architecture in general.
2. Consistent validation and dispatch rules: as all the reports pass the Business Layer before storing data, the Damage Processing Component can apply the same validation (duplicates check, mandatory fields, photo validation) and the same ticket dispatch rules to all the departments uniformly.